

• From this list, it's easy to see that none of the sums give a perfect cube less than 1000.

∴ There are no positive perfect cubes less than 1000 that are the sum of the cubes of two positive integers. $\square$

5. Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$.

Solution: There can be 3 possible cases:

Case 1: $x \geq y$. ∴ $\max(x, y) = x$ $\min(x, y) = y$.
∴ $\max(x, y) + \min(x, y) = x + y$.

Case 2: $x < y$. ∴ $\max(x, y) = y$ $\min(x, y) = x$
∴ $\max(x, y) + \min(x, y) = y + x = x + y$ (Commutative Law)

Case 3: $x = y$. ∴ $\max(x, y) = \min(x, y) = x = y$
 $\max(x, y) + \min(x, y) = x + y$

∴ $\max(x, y) + \min(x, y) = x + y$ $\square$

6. Use a proof by cases to show that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a, b and c are real numbers.

Solution: The following possible cases arise:

1. ~~$a \geq b \geq c$~~ $\min(b, c) = c$. ∴ $\min(a, \min(b, c)) = \min(a, c) = c$
 $\min(a, b) = b$ ∴ $\min(\min(a, b), c) = c$

This implies $\min(a, \min(b, c)) = \min(\min(a, b), c)$

2. $a \geq c \geq b$. $\min(b, c) = b$ $\min(a, \min(b, c)) = \min(a, b) = b$
 $\min(\min(a, b), c) = \min(b, c) = b$

∴ $\min(a, \min(b, c)) = \min(\min(a, b), c)$